One-Variable Statistics, Distributions & Margin of Error

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the botanical conservatory. What is the mean of the data set?

Answer: 13. From the frequency table, the values are $10, 12, 14, 16$ with frequencies $2, 4, 4, 2$. Because the distribution is symmetric about $\frac{12+14}{2} = 13$, the arithmetic mean is $\frac{10(2)+12(4)+14(4)+16(2)}{12} = 13$.

2 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the regional bakery. What is the total number of measurements in the data set?

Answer: 18. Summing the frequencies in the second column of the table gives $2 + 7 + 7 + 2 = 18$ total measurements.

3 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the science museum. What is the mean of the data set?

Answer: 25. From the frequency table, the values are $22, 24, 26, 28$ with frequencies $2, 6, 6, 2$. Because the distribution is symmetric about $\frac{24+26}{2} = 25$, the arithmetic mean is $\frac{22(2)+24(6)+26(6)+28(2)}{16} = 25$.

4 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the ceramic studio. What is the total number of measurements in the data set?

Answer: 14. Summing the frequencies in the second column of the table gives $2 + 5 + 5 + 2 = 14$ total measurements.

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5 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the public transit depot. What is the mean of the data set?

Answer: 37. From the frequency table, the values are $34, 36, 38, 40$ with frequencies $2, 4, 4, 2$. Because the distribution is symmetric about $\frac{36+38}{2} = 37$, the arithmetic mean is $\frac{34(2)+36(4)+38(4)+40(2)}{12} = 37$.

6 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the art gallery. What is the total number of measurements in the data set?

Answer: 18. Summing the frequencies in the second column of the table gives $2 + 7 + 7 + 2 = 18$ total measurements.

7 [Foundational] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the university lecture hall. What is the mean of the data set?

Answer: 49. From the frequency table, the values are $46, 48, 50, 52$ with frequencies $2, 6, 6, 2$. Because the distribution is symmetric about $\frac{48+50}{2} = 49$, the arithmetic mean is $\frac{46(2)+48(6)+50(6)+52(2)}{16} = 49$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the film archive, in a calculation for the orchestra hall (case 12), what is the mode of the data set: $44, 16, 10, 44, 45, 9, 12$?

Answer: 44. The value 44 appears twice, more often than any other value, so it is the mode.

9 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the mountain observatory, in a calculation for the honeybee apiary (case 16), what is the mode of the data set: $42, 1, 2, 18, 11, 8, 2, 41$?

Answer: 2. The value 2 appears twice, more often than any other value, so it is the mode.

10 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the recycling center, the numbers of books read by five students last month are: 6, 10, 14, 18, and 22. What is the mean number of books read?

Answer: 14. Sum = $6 + 10 + 14 + 18 + 22 = 70$. Count = 5. Mean = $70/5 = 14$.

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11 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the water treatment plant, the mean of four numbers is 18. Three of the numbers are 12, 16, and 22. What is the fourth number?

Answer: 22. Total sum = $4 \times 18 = 72$. Sum of three = $12 + 16 + 22 = 50$. Fourth number = $72 - 50 = 22$.

12 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the regional airport shuttle, in a calculation for the coastal aquarium (case 28), if five quiz scores have a sum of 420, what is the mean score?

Answer: 84. Applying One-Variable Statistics, Distributions & Margin of Error principles, Mean = $420/5 = 84$. Therefore, the verified correct option is 84.

13 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the glassblowing studio, in a class of 25 students, the median test score is 84. If 12 students scored above 84, how many students scored at or below 84?

Answer: 13. Total students = 25. If 12 scored above 84, then $25 - 12 = 13$ scored at or below 84.

14 [Intermediate] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the ice skating rink, in a calculation for the wildlife sanctuary (case 36), what is the mode of the values: $12, 15, 18, 15, 21, 15, 18$?

Answer: 15. 15 appears 3 times, which is more than any other value.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the ceramic studio, a data set of 9 numbers has a median of 15. If the number 15 is added to the data set, what is the new median?

Answer: 15. Adding a number equal to the current median leaves the median unchanged at 15.

16 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

During a 5-day study at the botanical conservatory, the mean number of adult tickets recorded per day was 20. For the first four days, the counts were $15, 18, 21,$ and 22. What was the count on the fifth day?

Answer: 24. For 5 values to have a mean of 20, their sum must equal $5 \times 20 = 100$. The sum of the first four values is 76, so the fifth value is $100 - 76 = 24$.

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17 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A random sample of visitors to the municipal library found that 36% prefer hardcover books, with an associated margin of error of $\pm 4\%$. If a total of 3000 visitors attend next month, what is the difference between the maximum and minimum plausible number of visitors expected to prefer hardcover books?

Answer: 240. The plausible interval for the percentage is from 32% to 40%, a span of 8%. Multiplying 3000 by $8\% = 0.08$ gives $1200 - 960 = 240$ visitors.

18 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

At the community theater, Data Set A consists of the 5 measurements $34, 38, 42, 46, 50$, and Data Set B consists of the 5 measurements $24, 33, 42, 51, 60$. What is the positive difference between the range of Data Set B and the range of Data Set A?

Answer: 20. The range of Data Set B is $60 - 24 = 36$, and the range of Data Set A is $50 - 34 = 16$. The positive difference is $36 - 16 = 20$.

19 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

During a 5-day study at the regional bakery, the mean number of standard passes recorded per day was 23. For the first four days, the counts were $18, 21, 24,$ and 25. What was the count on the fifth day?

Answer: 27. For 5 values to have a mean of 23, their sum must equal $5 \times 23 = 115$. The sum of the first four values is 88, so the fifth value is $115 - 88 = 27$.

20 [Challenging] • [SPR] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A random sample of visitors to the coastal aquarium found that 39% prefer oak saplings, with an associated margin of error of $\pm 4\%$. If a total of 6000 visitors attend next month, what is the difference between the maximum and minimum plausible number of visitors expected to prefer oak saplings?

Answer: 480. The plausible interval for the percentage is from 35% to 43%, a span of 8%. Multiplying 6000 by $8\% = 0.08$ gives $2580 - 2100 = 480$ visitors.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the municipal library. What is the total number of measurements in the data set?

a. 17

b. 15

c. 19

d. 16 ✓

Summing the frequencies in the second column of the table gives $3 + 5 + 5 + 3 = 16$ total measurements.

22 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the community theater. What is the mean of the data set?

a. 16

b. 20

c. 17 ✓

d. 18

From the frequency table, the values are $14, 16, 18, 20$ with frequencies $4, 6, 6, 4$. Because the distribution is symmetric about $\frac{16 + 18}{2} = 17$, the arithmetic mean is $\frac{14(4) + 16(6) + 18(6) + 20(4)}{20} = 17$.

23 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the coastal aquarium. What is the mean of the data set?

a. 21 ✓

b. 22

c. 20

d. 24

From the frequency table, the values are $18, 20, 22, 24$ with frequencies $3, 4, 4, 3$. Because the distribution is symmetric about $\frac{20 + 22}{2} = 21$, the arithmetic mean is $\frac{18(3) + 20(4) + 22(4) + 24(3)}{14} = 21$.

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24 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the mountain observatory. What is the total number of measurements in the data set?

a. 19

b. 17

c. 21

d. 18 ✓

Summing the frequencies in the second column of the table gives $4 + 5 + 5 + 4 = 18$ total measurements.

25 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the urban bike-share station. What is the total number of measurements in the data set?

a. 23

b. 20 ✓

c. 21

d. 19

Summing the frequencies in the second column of the table gives $3 + 7 + 7 + 3 = 20$ total measurements.

26 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the solar microgrid facility. What is the mean of the data set?

a. 29 ✓

b. 30

c. 28

d. 32

From the frequency table, the values are $26, 28, 30, 32$ with frequencies $4, 4, 4, 4$. Because the distribution is symmetric about $\frac{28 + 30}{2} = 29$, the arithmetic mean is $\frac{26(4) + 28(4) + 30(4) + 32(4)}{16} = 29$.

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27 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the agricultural cooperative. What is the mean of the data set?

a. 32

b. 36

c. 33 ✓

d. 34

From the frequency table, the values are $30, 32, 34, 36$ with frequencies $3, 6, 6, 3$. Because the distribution is symmetric about $\frac{32+34}{2} = 33$, the arithmetic mean is $\frac{30(3)+32(6)+34(6)+36(3)}{18} = 33$.

28 [Foundational] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

The frequency table shown summarizes the daily score measurements collected at the publishing house. What is the total number of measurements in the data set?

a. 25

b. 22 ✓

c. 23

d. 21

Summing the frequencies in the second column of the table gives $4 + 7 + 7 + 4 = 22$ total measurements.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the ice skating rink, a two-way table of $N = 127$ observations on commuting mode records cell counts $u_1 = 30$ for (Junior, Train) and $v_2 = 35$ for (Senior, Bus). What is the diagonal sum $s = u_1 + v_2$?

a. 63

b. 65 ✓

c. 71

d. 68

Summing the two frequencies gives $30 + 35 = 65$.

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30 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the meteorological station, a two-way table of $N = 88$ observations on commuting mode records cell counts $F_1 = 22$ for (Junior, Train) and $G_2 = 21$ for (Senior, Bus). What is the diagonal sum $w = F_1 + G_2$?

a. 41

b. 49

c. 46

d. 43 ✓

Summing the two frequencies gives $22 + 21 = 43$.

31 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the municipal library, in a calculation for the textile workshop (case 14), what is the range of the data set: $98, 30, 94, 76, 10, 11, 28$?

a. 88 ✓

b. 80

c. 95

d. 82

Applying One-Variable Statistics, Distributions & Margin of Error principles, Range $= \max - \min = 98 - 10 = 88$ Therefore, the verified correct option is 88.

32 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the regional bakery, in a calculation for the geology lab (case 15), what is the mean of the data set: $7, 40, 31, 18$?

a. 23

b. 24 ✓

c. 21

d. 32

Mean $= \frac{7 + 40 + 31 + 18}{4} = \frac{96}{4} = 24$

33 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the urban bike-share station, in a calculation for the carpentry workshop (case 17), what is the median of the data set: $45, 41, 59, 13, 65, 54, 68$?

a. 54 ✓

b. 57

c. 48

d. 45

Ordered: $13, 41, 45, 54, 59, 65, 68$. The middle value is 54.

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34 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the ceramic studio, in a calculation for the historic lighthouse (case 18), what is the mode of the data set: $33, 34, 27, 3, 4, 34$?

a. 40

b. 32

c. 34 ✓

d. 42

The value 34 appears twice, more often than any other value, so it is the mode.

35 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the publishing house, data set A: $44, 47, 50$. Data set B: $35, 47, 59$. Which data set has the greater standard deviation?

a. Data set A

b. **Data set B** ✓

c. They are equal

d. Cannot be determined

Data set A spans from 44 to 50; data set B spans from 35 to 59. The one with greater spread from the mean has the greater standard deviation: Data set B.

36 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the art gallery, what is the median of the following list of numbers: $3, 7, 12, 15, 19, 24, 30$?

a. 12

b. 15 ✓

c. 17

d. 19

The ordered list has 7 values. The middle (4th) value is 15.

37 [Intermediate] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the robotics club, a biologist recorded the weights of six laboratory mice in grams: 14, 28, 19, 45, 12, and 33. What is the range of these weights?

a. 27

b. 31

c. 33 ✓

d. 45

Range = Maximum - Minimum = $45 - 12 = 33$.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the mountain observatory, a researcher wants to study the sleeping habits of all college students in a state. Which sample is most representative?

a. 50 students selected at random from the university library at midnight

b. 100 student athletes from one college

c. **500 students selected at random from across all colleges and universities in the state ✓**

d. 200 students who volunteer to fill out an online sleep survey

Random selection across the entire population of interest (all colleges in the state) provides the most representative sample.

39 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the urban bike-share station, what happens to the standard deviation of a data set if 10 is added to every data point?

a. It increases by 10.

b. It increases by $\sqrt{10}$.

c. **It remains unchanged. ✓**

d. It is multiplied by 10.

Adding a constant shifts the distribution along the number line without altering the spread between data points, so standard deviation is unchanged.

40 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the publishing house, a study randomly assigns 100 headache patients to receive either a new medication or a placebo. What type of study is this?

a. Observational study

b. **Randomized controlled experiment ✓**

c. Census

d. Sample survey

Treatments are actively imposed and participants are randomly assigned, making it a randomized controlled experiment.

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41 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

In a model for the recycling center, in a calculation for the local farmers market (case 70), why is random assignment used in experiments?

a. To ensure the sample represents the entire global population

b. **To balance confounding variables across treatment groups so causality can be established** ✓

c. To eliminate the need for a control group

d. To reduce the sample size needed

Random assignment distributes confounding variables evenly between groups, allowing researchers to infer cause-and-effect relationships.

42 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

At the wildlife sanctuary, Data Set A consists of the 5 measurements $52, 56, 60, 64, 68$, and Data Set B consists of the 5 measurements $46, 53, 60, 67, 74$. What is the positive difference between the range of Data Set B and the range of Data Set A?

a. 12 ✓

b. 3

c. 28

d. 16

The range of Data Set B is $74 - 46 = 28$, and the range of Data Set A is $68 - 52 = 16$. The positive difference is $28 - 16 = 12$.

43 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

During a 5-day study at the water treatment plant, the mean number of fiction novels recorded per day was 41. For the first four days, the counts were $36, 39, 42,$ and 43. What was the count on the fifth day?

a. 41

b. 43

c. 48

d. 45 ✓

For 5 values to have a mean of 41, their sum must equal $5 \times 41 = 205$. The sum of the first four values is 160, so the fifth value is $205 - 160 = 45$.

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44 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

A random sample of visitors to the orchestra hall found that 37% prefer tulip bulbs, with an associated margin of error of $\pm 4\%$. If a total of 3000 visitors attend next month, what is the difference between the maximum and minimum plausible number of visitors expected to prefer tulip bulbs?

a. 1110

b. 290

c. 240 ✓

d. 120

The plausible interval for the percentage is from 33% to 41%, a span of 8%. Multiplying 3000 by $8\% = 0.08$ gives $1230 - 990 = 240$ visitors.

45 [Challenging] • [MCQ] • Target: Mean, Median, Range, Standard Deviation & Margin of Error

At the sports complex, Data Set A consists of the 5 measurements $55, 59, 63, 67, 71$, and Data Set B consists of the 5 measurements $43, 53, 63, 73, 83$. What is the positive difference between the range of Data Set B and the range of Data Set A?

a. 16

b. 24 ✓

c. 6

d. 40

The range of Data Set B is $83 - 43 = 40$, and the range of Data Set A is $71 - 55 = 16$. The positive difference is $40 - 16 = 24$.

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